

TODD TAKALA

DATA SCIENTIST, MACHINE LEARNING & AI

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PROFESSIONAL SUMMARY

Data scientist with 25+ years in capital equipment engineering, from the shop floor as a mechanic and welder to enterprise machine learning. Ships production ML for predictive maintenance, forecasting, anomaly detection, and LLM applications built on failure data, IoT telemetry, and ERP systems, and owns the full MLOps lifecycle on AWS and Snowflake. Caterpillar CEO Award recipient and 2x Amazon ML competition champion.

PROFESSIONAL EXPERIENCE

Ford Motor Company

Sep 2026 – Present

Senior Data Scientist, Manufacturing Intelligence, Machine Learning & AI

- Architect graph network ontology of automotive manufacturing plant equipment within Neo4j and Google Cloud Platform.

Stack: Python, GCP, Neo4j, Docker, Make

Caterpillar Inc.

Sep 2022 – Jun 2026

Lead Data Scientist

- Designed and deployed a TensorFlow ordinal deep-learning model predicting equipment repair times, using sentence transformer embeddings of labor descriptions across 2M+ combinations of labor code, sales model, and build variant, which improved accuracy to 98% from 54%, tripled warranty coverage, and cut \$4M annually in time studies.
- Engineered the training labels rather than inheriting them: found warranty labor times were censored at the published standard, excluded them, and built ground truth from time studies, engineering estimates, and outlier trimmed non-warranty medians.
- Governed the model with an explicit abstention rule (no prediction on unseen build variants or job codes) plus per-segment validation gates at a 90% floor; still delivered 95% coverage of warranty labor lines against an 80% business target.
- Owned the full MLOps lifecycle on AWS SageMaker (Infrastructure as Code, CI/CD, Docker, drift detection, and automated retraining) and architected the Snowflake data foundation behind the team's analytics platform.
- Delivered production ML across tabular, time series, and LLM applications for warranty adjudication, insurance reserving, and service quoting; mentored early-career data scientists on the team.

Stack: Python, R, SQL · PyTorch, TensorFlow/Keras, XGBoost, scikit-learn · Snowflake · AWS (SageMaker, Glue, Athena, Lambda, Step Functions, ECS/EKS, CloudWatch) · Docker · Azure DevOps, GitHub Actions

Amazon.com Tempe, AZ

Feb 2022 – Sep 2022

Reliability Analytics Manager

- Replaced compliance-dependent failure reporting with a machine-generated signal: built a pipeline consuming hourly serialized component logs from S3 and reconstructed true component life from serial-number changes, then moved the fleet design standard from average life to B10.
- Applied Weibull and survival analysis across a global fleet of 400,000 autonomous robots, informing maintenance strategy and design metrics network-wide.
- Led a team of 5 BI engineers building a global robot monitoring pipeline (R/Shiny, AWS) adopted across sites for incident response and proactive maintenance.

Stack: R, Python · scikit-learn, AutoGluon · Shiny · PostgreSQL · Docker · AWS (Redshift, Athena, SageMaker, S3)

Empire Cat Mesa, AZ

Jan 2012 – Feb 2022

Reliability Engineering Manager & Data Engineering Manager

Dec 2016 – Feb 2022

- Led modeling on a 25-person cross-functional program that replaced Caterpillar's estimated engine life target with an evidence based one: reduced thousands of candidate factors via PCA to 9 controllable variables, then built a diesel engine digital twin on live telemetry to monitor them, extending haul truck engine life 23% and saving the client \$62M in 90 days.
- Connected reliability Weibull models to parts demand, which reduced part stock outs by 25%.
- Architected a component exchange program with data warehousing, ETL/ELT pipelines, and time series models to target at-risk components, with client-facing ownership of the reliability and product problem management programs.
- Established data engineering governance, quality standards, and a metadata framework adopted across the dealership.

Stack: SQL Server, T-SQL · R (ggplot2, tidyverse, RMarkdown) · Python (Pandas, scikit-learn) · SQLite · Jupyter · Docker · Ubuntu, Linux

Lead Reliability Engineer

Jun 2016 – Dec 2016

- Led root cause failure analysis on an aging mining truck line, remediating engine, drivetrain, and hydraulic failures to raise availability from 80% to 90%.
- Applied design of experiments and statistical modeling to isolate unreliability drivers; created an RMarkdown research framework Caterpillar incorporated into its CRM.

Technical Communications Manager (Engineering Manager)

Dec 2012 – Jun 2016

- Engineering manager of a 5-person team delivering technical communications, reliability analytics, and data engineering for a 250-truck mining fleet.
- Applied DMAIC Six Sigma and data science to identify and prioritize reliability improvements across a fleet-wide program.
- Developed a fluids and wear analysis application (R, SQL) using oil tribology and spectroscopy to drive proactive maintenance, eliminating \$100K/year in license costs.

Reliability Engineer

Jan 2012 – Dec 2012

- Led the Product Problem Management program, applying 8D and RCFA to resolve recurring failures and reduce downtime.
- Analyzed field failure data to prioritize the highest-impact failure modes and guide remediation planning.

Road Machinery, LLC *Phoenix, AZ*

Jan 2008 – Jan 2012

Technical Communicator (Engineer)

- Direct engineering support to clients and field service technicians, including root cause failure analysis, maintenance guidance, and life cycle cost analysis.
- Built the Total Cost of Ownership analysis underpinning a \$72M sale of electric mining haul trucks; principal consultant on a \$2.5M gold mine account in Mexico.

Stack: SQL Server, T-SQL · MATLAB/Octave · SolidWorks, finite element analysis · LaTeX

Komatsu North America *Chattanooga, TN*

Jun 2006 – Dec 2007

Design Engineer

- Led end-to-end development of a 25-ton hydraulic rescue winch from concept through prototyping, field testing, and release.
- Ran FMEA and FRACAS on prototype and production machines, authoring corrective actions that drove design changes and improved durability before release.

Stack: MATLAB/Octave · PTC · finite element analysis

EARLY CAREER

Engineering Intern, Xtek Mining Services (2005) · Heavy Equipment Operator and Mechanic, Ryan Incorporated Central, Roland Machinery Co., and Harry W. Kuhn, Inc. (1997–2002), diagnosing and repairing mechanical, hydraulic, electrical, and control system failures on heavy capital equipment.

EDUCATION

Bachelor of Science in Mechanical Engineering, Arizona State University

2006

Certificate of Apprenticeship, Operating Engineer, U.S. Department of Labor & I.U.O.E. Local 150 AISP

2001

SELECTED CERTIFICATIONS

AWS Certified Machine Learning – Specialty (MLS-C01) · Lean Six Sigma Black Belt, Kennesaw State University · Certified SAFe Agilist · Advanced Deep Learning Specialist, IBM · Caterpillar Applied Failure Analysis

RECOGNITION & SPEAKING

Caterpillar CEO Award (2024): AI-driven labor time model, for exceptional personal contribution and business impact.

2x ML Competition Champion, Amazon (2022): Internal regression and recommender system competitions.

Invited Speaker, Data Science for Fleet Reliability (2021) at Caterpillar Six Sigma List Manager Workshop, Peoria, IL: Diesel engine digital twin on live telemetry, tracking 9 PCA-derived controllable variables to set an evidence based engine life target.

Invited Speaker, Reliability & Rebuild Programs (2015, 2019) at Caterpillar, Tinaja Proving Grounds: Product Problem Management; and a truck rebuild program at Caterpillar's large mining truck conference, delivering 2–4% higher availability at 60% of new-machine cost.

CORE COMPETENCIES

Machine Learning & AI: Deep learning (PyTorch, TensorFlow/Keras) · time series forecasting · anomaly detection · survival analysis and censored data · ordinal regression · NLP and embeddings · design of experiments

MLOps, Data & Cloud: AWS SageMaker · Snowflake · CI/CD and Infrastructure as Code · Docker · model governance, validation gates, and drift detection · automated retraining · ETL/ELT pipelines · data architecture and governance · Linux

Reliability Engineering: Root cause failure analysis · 8D and 5 Whys · FMEA / DFMEA / PFMEA · FRACAS · Weibull analysis · prognostics and remaining useful life · condition monitoring · oil tribology and spectroscopy · life cycle cost analysis

Domain Expertise: Heavy equipment, mining, automotive and manufacturing · fleet and maintenance management · warranty and service operations · supply chain and demand planning